

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [3]: df=pd.read_csv("Social_Network.csv ")
df
```

```
Out[3]:
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [4]: df.isnull()
```

Out[4]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
395	False	False	False	False	False
396	False	False	False	False	False
397	False	False	False	False	False
398	False	False	False	False	False
399	False	False	False	False	False

400 rows × 5 columns

In [5]: `df.count()`

```
Out[5]: User ID      400
        Gender      400
        Age         400
        EstimatedSalary  400
        Purchased    400
        dtype: int64
```

In [6]: `df.duplicated()`

```
Out[6]: 0      False
        1      False
        2      False
        3      False
        4      False
        ...
        395    False
        396    False
        397    False
        398    False
        399    False
        Length: 400, dtype: bool
```

```
In [7]: df.drop(columns=["User ID"], inplace=True)
df["Gender"]=df["Gender"].map({"Male":1,"Female":0})
df
```

```
Out[7]:
```

	Gender	Age	EstimatedSalary	Purchased
0	1	19	19000	0
1	1	35	20000	0
2	0	26	43000	0
3	0	27	57000	0
4	1	19	76000	0
...	...	...	...	...
395	0	46	41000	1
396	1	51	23000	1
397	0	50	20000	1
398	1	36	33000	0
399	0	49	36000	1

400 rows × 4 columns

```
In [8]: x=df[["Gender","Age","EstimatedSalary"]]
        y=df["Purchased"]
```

```
In [9]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
In [10]: scaler=StandardScaler()
         x_train=scaler.fit_transform(x_train)
         x_test=scaler.transform(x_test)
```

```
In [11]: model=LogisticRegression()
         model.fit(x_train,y_train)
```

```
Out[11]: LogisticRegression ⓘ ?
         ► Parameters
         ► Fitted attributes
```



```
In [12]: y_pred=model.predict(x_test)
```

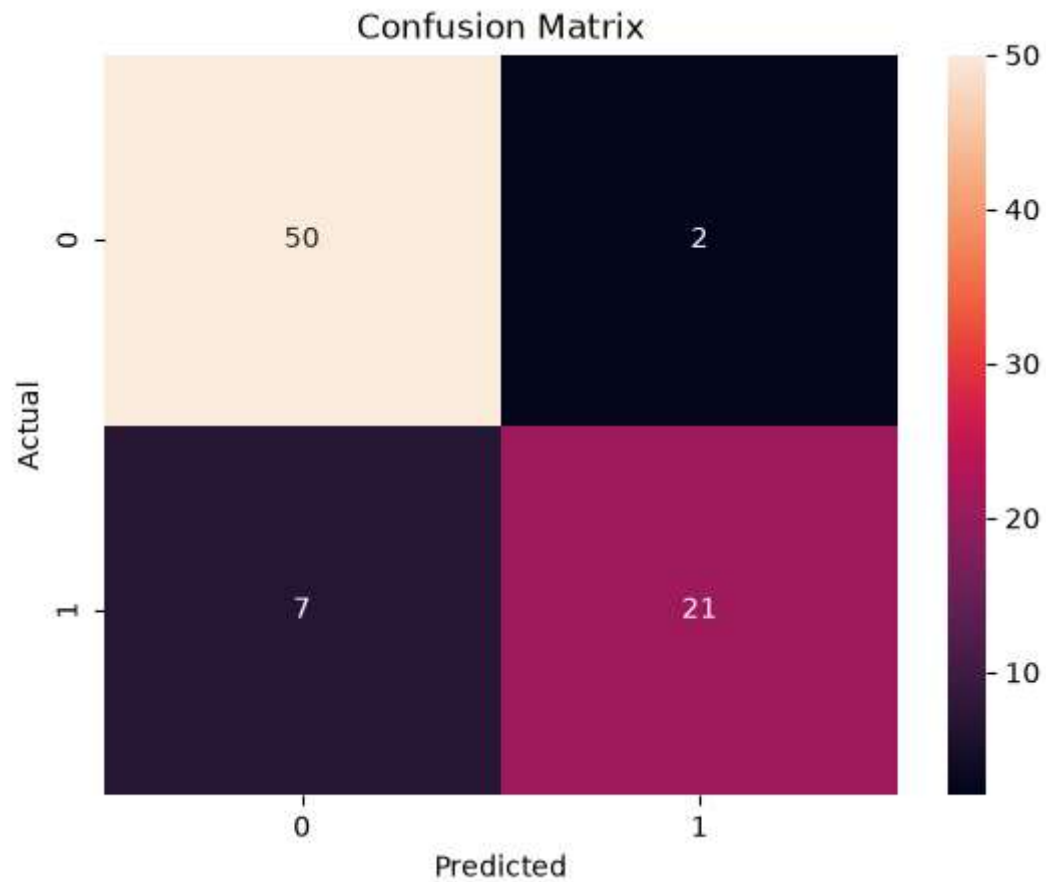
```
In [13]: accuracy=accuracy_score(y_test,y_pred)
         accuracy
```

```
Out[13]: 0.8875
```

```
In [14]: conf_matrix=confusion_matrix(y_test,y_pred)
         conf_matrix
```

```
Out[14]: array([[50,  2],
               [ 7, 21]])
```

```
In [15]: sns.heatmap(conf_matrix,annot=True)
         plt.xlabel("Predicted")
         plt.ylabel("Actual")
         plt.title("Confusion Matrix")
         plt.show()
```



```
In [21]: new_data = np.array([[1, 30, 80000]])  
  
new_data_df = pd.DataFrame(new_data, columns=x.columns)  
  
new_data_scaled = scaler.transform(new_data_df)  
  
predicted = model.predict(new_data_scaled)  
  
if predicted == 1:  
    print("Purchase : Yes")  
else:  
    print("Purchase : No")
```

Purchase : No

